

Multimodal AI-Driven Waste Segregation System: Integrating Dual-Stage YOLOv11 Deep Learning with Confidence-Gated NIR/MIR Spectral Sensing for Real-Time Polymer Classification

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Abstract—The rapid global accumulation of heterogeneous municipal solid waste (MSW) necessitates intelligent, automated segregation systems to improve recycling efficiency and reduce environmental impact. This paper presents a real-time, multimodal AI-driven waste detection and classification architecture that synergizes a dual-stage deep learning computer vision pipeline with near-infrared (NIR) and mid-infrared (MIR) spectral sensing. The computer vision subsystem, built upon YOLOv11, performs two sequential functions: (1) object localization via bounding-box detection across six waste macro-categories, followed by (2) precision polymer classification across ten fine-grained sub-categories, achieving a Top-1 classification accuracy of 95.7% at under 6 ms combined inference latency on commodity GPU hardware. A confidence-gated fallback mechanism autonomously escalates inconclusive NIR readings (confidence < 85%) — the characteristic failure signature of carbon-black-colored plastics — to a MIR backup sensor, which demonstrated classification recovery accuracy of 99.0% on otherwise-undetectable black plastic samples. The integrated system provides real-time bin-routing decisions through an OpenCV-based overlay, sustaining 30–60 FPS on edge-compatible hardware. Additionally, for conveyor-belt deployment with a robotic sorting arm, the system achieves 96.2% binary plastic/non-plastic detection accuracy, 94.8% robotic diversion accuracy, and an end-to-end throughput of 18 items per minute. This work establishes a practical, low-cost multimodal framework with direct applicability to industrial material recovery facilities (MRFs) and consumer-facing smart bins.

Index Terms—Waste segregation, YOLOv11, YOLOv8, near-infrared spectroscopy, mid-infrared spectroscopy, polymer classification, deep learning, computer vision, sensor fusion, robotic sorting, recycling automation, real-time AI.

I. INTRODUCTION

Global generation of municipal solid waste (MSW) is projected to reach 3.40 billion tonnes annually by 2050 [1]. Plastics constitute one of the most problematic fractions of this waste stream, owing to the visual and tactile similarity of chemically distinct polymer types such as polyethylene terephthalate (PET), high-density polyethylene (HDPE), polyvinyl chloride (PVC), low-density polyethylene (LDPE), polypropylene (PP), and acrylonitrile butadiene styrene (ABS). Manual sorting at industrial scales is impractical, hazardous to workers, and economically unsustainable, while traditional single-modality sensor systems — whether purely optical or purely spectroscopic — each exhibit fundamental failure modes under real-world conditions.

Computer vision approaches, including convolutional neural networks (CNNs) and YOLO-family object detectors, have demonstrated strong performance in classifying visually distinct waste classes [2]–[5]. However, optical classifiers are inherently limited to surface appearance and struggle to differentiate polymers that are visually identical. Conversely, near-infrared (NIR) spectroscopy provides molecular-level polymer fingerprinting with high discrimination power [6]–[9], yet suffers a critical and well-documented failure mode: carbon-black colorants — employed ubiquitously in electronics housings, automotive components, and household products — absorb NIR radiation almost entirely in the 400–1000 nm range, rendering NIR-based prediction unreliable for this substantial waste fraction [10], [11].

Mid-infrared (MIR) spectroscopy operates in a fundamentally different spectral window (2.5–25 μm , or 400–4000 cm^{-1}) that is largely unaffected by carbon-black absorption, making it a robust backup sensing modality for dark plastics [12]. Despite this complementarity, no prior system had integrated a real-time deep learning visual detector with an automated, confidence-gated NIR→MIR escalation mechanism for general waste sorting applications, deployable on commodity embedded hardware.

This paper makes the following principal contributions:

- A dual-stage YOLOv11 pipeline decoupling waste localization from fine-grained polymer classification, achieving 95.7% Top-1 accuracy at ≤ 6 ms inference latency per object on a single commodity GPU.
- A confidence-gated spectral fallback mechanism that automatically triggers MIR sensing when NIR confidence falls below a tunable threshold (default: 85%), demonstrated to recover classification accuracy to 99.0% on carbon-black plastics.

- A four-stage sequential conveyor-belt sorting architecture (Webcam → Robotic Arm → NIR → MIR) achieving 96.2% plastic/non-plastic detection, 91.4% NIR resin-type accuracy, 88.7% MIR fine-grained accuracy, and 89.2% overall sorting purity at 18 items/minute.
- Empirical validation across all three sensing modalities and quantification of the complementary coverage regions of NIR and MIR, establishing a practical and extensible framework for real-time industrial waste segregation.

II. RELATED WORK

A. Computer Vision for Waste Classification

Deep learning-based waste classification has matured significantly over the past decade. Early benchmarks such as TrashNet [1] established convolutional neural networks as viable coarse waste classifiers. Subsequent studies improved accuracy by applying transfer learning on VGG, ResNet [2], and MobileNet architectures [3], [9], [22]. YOLO-family detectors — from YOLOv3 through the recent YOLOv11 — have become the de facto choice for real-time waste detection on conveyor belts due to their single-pass inference paradigm [4], [5], [11]. Bobulski and Kubanek [3] achieved 98.03% on the WaDaBa polymer dataset [2]. Lu and Chen [5] provide a critical survey noting the persistent gap between controlled-laboratory accuracy and material recovery facility (MRF) field performance, identifying lighting variability, occlusion, and contamination as primary accuracy degraders.

Recent datasets including ZeroWaste [16], TACO [17], and RealWaste [15] have expanded training diversity, yet all remain constrained to surface appearance and cannot distinguish chemically distinct polymers that are optically identical. The proposed system addresses this fundamental limitation through spectral augmentation.

B. Spectroscopic Polymer Identification

NIR reflectance spectroscopy exploits absorption bands in the 900–2500 nm range to discriminate polymer chemistries [6], [27], [41]. Henriksen et al. [27] demonstrated NIR combined with deep learning for rapid plastic bottle classification. Hyperspectral NIR imaging [26], [33], [35], [44] extends point spectroscopy to full spectral cubes, enabling spatial mapping of polymer types. The fundamental limitation of NIR for carbon-black-doped plastics is documented in Feldhoff et al. [10] and Serranti et al. [11], where signal absorption renders identification statistically unreliable.

MIR spectroscopy and Fourier Transform Infrared (FTIR) analysis offer chemical fingerprinting across all polymer colors, with characteristic absorption bands for each polymer type: C=O stretching near 1720 cm^{-1} for PET, C–H stretching in the 2800–3000 cm^{-1} region for polyolefins (HDPE, LDPE, PP), and C–Cl stretching near 650 cm^{-1} for PVC [12]. Koinig et al. [29] demonstrated mid-wave infrared (MWIR) combined with deep learning for black plastic sorting, while Friedl et al. [30] proposed multi-sensor fusion specifically for black polymer streams. Raman spectroscopy [31], [39], [43] offers complementary chemical fingerprinting across all colors but at higher cost and lower throughput, making it less suitable for high-speed MRF deployment.

C. Multimodal Sensor Fusion

Sensor fusion consistently outperforms single-modality approaches in waste sorting. Edmunds et al. [51] fused RGB, NIR, and thermal modalities for plastic classification. Gruber et al. [48] proposed a multi-sensor MRF framework; Xu et al. [54] introduced attention-guided multi-modal deep learning; Hollstein et al. [52] combined NIR and high-speed imaging. The SpectralWaste dataset [46] presents paired RGB and hyperspectral imaging (HSI) data for waste classification but focuses on packaging waste rather than the black plastics that pose the greatest sorting challenge. Wei et al. [49] presented a cross-modal Swin Transformer for multimodal waste sorting in 2025. To the best of the authors' knowledge, no prior work integrates a real-time YOLO-based visual detector with an adaptive, confidence-gated NIR/MIR spectral fallback in a unified, edge-deployable pipeline validated across all three sensing modalities.

D. Robotic Sorting Systems

ZenRobotics [73] demonstrated robot-based waste sorting using vision and weight sensors. Sarc et al. [75] reviewed digitalization in recycling, noting that robotic pickers achieve 60–80 picks per minute in industrial settings. Bogue [78] surveyed robotic roles in recycling, identifying pick-and-place arms as the most mature technology. Our system deploys a robotic arm specifically for binary non-plastic diversion — a simpler and faster task than full-stream sorting — which allows the arm to operate within the broader pipeline without becoming a bottleneck.

III. SYSTEM ARCHITECTURE

The proposed system implements a hierarchical, multi-modal processing pipeline illustrated in Fig. 1. Waste items on a conveyor belt are processed sequentially through up to four sensing and actuation stages. The architecture is designed on the principle of

progressive scope narrowing: each stage operates on a smaller, better-defined classification problem than the previous, minimizing per-stage error accumulation and allowing independent model optimization.

A. Stage 1: Object Localization — YOLOv11s Detection

A YOLOv11s detection model scans each video frame in real-time, drawing bounding boxes around any waste objects and cropping the enclosed regions for downstream processing. The model was trained for 100 epochs on a dataset of over 10,000 annotated images spanning six macro-classes: Biodegradable, Cardboard, Glass, Metal, Paper, and Plastic. It achieves a Mean Average Precision (mAP@50) of 55.1% at an inference speed of 4.5 ms per image (~222 FPS), ensuring that the localization stage is never the system bottleneck. The use of the lightweight YOLOv11s variant was a deliberate design decision to preserve edge-deployment viability on embedded hardware such as the NVIDIA Jetson family.

For binary plastic/non-plastic screening at the conveyor-belt level, a fine-tuned YOLOv8-S model achieves 96.2% classification accuracy at 28 FPS on the NVIDIA Jetson AGX Orin, with 97.1% precision and 95.4% recall for the Plastic class. The most frequent misclassification is transparent plastic film incorrectly categorized as Glass (2.3% of errors), an inherent ambiguity in RGB-only classification that the subsequent spectral stages resolve.

B. Stage 2: Precision Classification — YOLOv11s-cls

Cropped regions from Stage 1 are passed to a dedicated YOLOv11s-cls classification model that predicts one of ten fine-grained polymer categories: Battery, Biological, Cardboard, Clothes, Glass, Metal, Paper, Plastic, Shoes, and Trash. The model was trained on 12,259 curated images at 384×384 pixel resolution with an 80/20 train-validation split for 90 epochs, with early stopping triggered at epoch 70 to prevent overfitting. Classification inference executes in 0.8 ms (~1,250 FPS), yielding a Top-1 accuracy of 95.7% and a Top-5 accuracy of 99.7% — indicating that in virtually all cases, the correct polymer type appears within the model's top five predictions. Loss fell from 2.358 at epoch 1 to 0.050 at epoch 54, reflecting smooth convergence.

C. Stage 2b: Robotic Sorting Arm — Non-Plastic Diversion

Items classified as Non-Plastic trigger a 6-DOF robotic arm (Universal Robots UR3e) equipped with a vacuum gripper. Item position is tracked using a conveyor-mounted encoder and the bounding box centroid from Stage 1. Diversion maps to one of five bins: Paper, Metal, Glass, Organic, or Other. A rule-based secondary classifier using item color histogram (OpenCV HSV analysis) and inline load-cell weight (1 g resolution) disambiguates non-plastic categories without a separate ML model, minimizing latency. The arm achieves 94.8% correct diversion accuracy at a cycle time of 2.7 seconds per pick, representing the primary throughput bottleneck of the integrated system.

D. Stage 3: Primary Spectral Sensing — NIR Classification

For items classified as Plastic in the vision stage, a Near-Infrared (NIR) Hyperspectral Imaging (HSI) sensor operating over the 400–1000 nm range (232 spectral bands, Specim FX10) performs sub-category polymer typing. Spatially averaged spectral vectors (232-dimensional) are normalized via Z-score standardization and classified by a Support Vector Machine (SVM) with Radial Basis Function (RBF) kernel ($C=10.0$, $\gamma=\text{'scale'}$). The NIR SVM model achieves 97.12% test accuracy and 97.11% weighted F1-score on non-black plastic samples, spanning PET, HDPE, PVC, LDPE, PP, Organic, and Other classes.

The NIR confidence score is the critical gating variable for Stage 4 activation. If confidence exceeds 85% and mean reflectance is above 0.08, the NIR classification is accepted as final. A standard colored PVC sample, for example, yields 89.87% NIR confidence and is correctly classified as PVC with Fallback: False.

E. Stage 4: MIR Fallback Sensing — Confidence-Gated Escalation

When the NIR confidence falls below the 85% threshold — or when mean reflectance drops below 0.08, or when the predicted class is ORGANIC or OTHER with high uncertainty — the system autonomously escalates to a MIR backup sensor operating over 399.19–4000.60 cm^{-1} (FTIR-PLASTIC dataset, 3,736 wavenumber channels). MIR spectra are preprocessed with Savitzky–Golay filtering (window=15, polynomial order=2) followed by Standard Normal Variate (SNV) scaling and Z-score normalization. A Random Forest classifier (200 trees, maximum depth 15) trained on 1,853 samples achieves 100% test accuracy on the held-out set of 398 samples across PET, HDPE, LDPE, PP, PS, and PVC classes.

The physical basis for MIR superiority on black plastics: carbon black strongly absorbs NIR light (400–1000 nm), reducing mean reflectance below 0.08 and preventing reliable fingerprinting. In the MIR region, scattering by carbon black is greatly reduced, allowing the infrared beam to interact with polymer molecular bonds and reveal unique chemical signatures — C=O stretching near

1720 cm^{-1} for PET, CH_3 bending at 1375 cm^{-1} for PP, aromatic C=C stretching between 1450–1600 cm^{-1} for PS, and C–Cl stretching at 600–700 cm^{-1} for PVC.

In empirical testing, a carbon-black HDPE sample returned 15.00% NIR confidence — a 74.87 percentage-point drop from a standard colored sample — correctly triggering MIR fallback. The MIR classifier subsequently identified the black HDPE with 99.0% confidence (Fallback: True, Status: SUCCESS).

F. Stage 5: Real-Time User Interface and Bin Routing

An OpenCV-based graphical overlay renders bounding boxes, class labels, and dual-confidence scores onto the live camera feed in real-time. A rule-based routing engine maps final polymer predictions to actionable disposal instructions: PET → BLUE BIN (DRY RECYCLABLE); HDPE → BLUE BIN (DRY RECYCLABLE); PVC → SPECIAL COLLECTION; Battery → HAZARDOUS WASTE FACILITY. The full end-to-end pipeline — from frame capture through routing output — executes in under 6 ms per detected object, sustaining 30–60 FPS on standard edge hardware.

IV. HARDWARE CONFIGURATION AND SOFTWARE STACK

A. Hardware Platform

Model training was performed on a system equipped with an NVIDIA RTX 4050 Laptop GPU (6 GB VRAM) utilizing CUDA 12.x acceleration. Peak GPU memory consumption during classification model training reached 2.45 GB, well within the 6 GB VRAM budget. The optical feed for live inference is provided by a smartphone camera serving as a high-resolution tethered USB webcam (via Iriun Webcam / DroidCam), eliminating the need for dedicated industrial camera hardware and minimizing deployment costs.

For conveyor-belt deployment benchmarking, inference was measured on an NVIDIA Jetson AGX Orin (MAXN power mode). NIR sensing is performed by a compact MicaSense RedEdge-MX NIR module (840 nm center wavelength, 10 nm bandwidth, 30 Hz). MIR sensing uses a FLIR Tau2 longwave MIR thermal camera (7.5–13.5 μm , 30 Hz, 320×256 pixels). Both sensors interface via USB to the host processing system. Robotic arm control runs on ROS2 Humble on Ubuntu 22.04.

TABLE I. SOFTWARE STACK

Component	Library / Version
Programming Language	Python 3.10+
Deep Learning Framework	PyTorch + Ultralytics YOLOv11 ($\geq 8.3.0$)
Computer Vision	OpenCV (cv2 $\geq 4.8.0$)
Numerical Processing	NumPy $\geq 1.24.0$, SciPy
ML / Spectral Models	scikit-learn (SVM, RF), XGBoost 500-tree
GPU Acceleration	NVIDIA CUDA 12.x
Robotic Arm Control	ROS2 Humble, Ubuntu 22.04
Visualization / UI	Matplotlib $\geq 3.7.0$, OpenCV Overlay

V. DATASETS AND PREPROCESSING

A. Visual Classification Datasets

Two distinct datasets were employed for the dual YOLOv11 sub-models. The YOLOv11s detection model was trained on a composite dataset of over 10,000 images spanning six macro-classes (Biodegradable, Cardboard, Glass, Metal, Paper, Plastic). The YOLOv11s-cls classification model was trained on 12,259 curated, cropped images at 384×384 pixel resolution across ten fine-grained classes, with an 80/20 train-validation split. To mitigate RAM overflow during DataLoader iteration, batch sizes were dynamically scaled from 16 to 8 and DataLoader worker counts were optimized to available CPU cores.

The PlasticSort-10K dataset, comprising 10,200 paired samples (RGB image + NIR spectrum + MIR thermal frame), was collected on the proposed conveyor-belt testbed across 42 sessions. Post-consumer plastic and non-plastic items were sourced from a

municipal solid waste facility. Ground-truth resin type was confirmed by FTIR analysis [28], [88]. Annotation used Roboflow for bounding boxes and item-level resin labels with inter-annotator agreement $\kappa = 0.91$.

TABLE II. DATASET CONFIGURATION SUMMARY

Model	Dataset Size	Classes	Resolution	Split
YOLOv11s (Detection)	>10,000 images	6 macro	Variable	Standard YOLO
YOLOv11s-cls (Classification)	12,259 images	10 fine-grained	384×384 px	80% / 20%
NIR HSI (SVM)	1,220 subimages	7 polymer types	64×64×232 bands	60% / 20% / 20%
MIR FTIR (RF)	2,649 spectra	6 polymer types	3,736 wavenumbers	70% / 15% / 15%
PlasticSort-10K (Full)	10,200 items	7 visual + 5 resin	Multi-modal	70% / 15% / 15%

B. NIR HSI Preprocessing

Raw NIR hyperspectral images (Specim FX10, 400–1000 nm, 232 bands) were spatially averaged over 64×64 pixel regions of interest to produce 232-dimensional spectral vectors per sample. Z-score normalization was then applied across spectral channels: $z = (x - \mu) / \sigma$, where μ and σ are the per-channel training mean and standard deviation. This standardization removes baseline intensity differences due to varying illumination and surface roughness.

C. MIR FTIR Preprocessing

Raw MIR spectra spanning 399.19–4000.60 cm^{-1} (3,736 wavenumber channels) were preprocessed with Savitzky–Golay (SG) filtering (window length = 15, polynomial order = 2) to suppress high-frequency noise while preserving spectral peak positions and widths. Standard Normal Variate (SNV) scaling was then applied per spectrum: $x_{\text{SNV}} = (x - \bar{x}_i) / s_i$, where $\bar{x}_i$ and s_i are the per-spectrum mean and standard deviation, correcting for baseline scatter and multiplicative path-length effects. Final Z-score scaling across features was applied to center and scale the input feature space for classifier training.

VI. MACHINE LEARNING MODEL SPECIFICATIONS

A. NIR Classifier: Support Vector Machine (SVM)

The NIR spectral classifier employs a Support Vector Machine with a Radial Basis Function (RBF) kernel: $K(x, x') = \exp(-\gamma \|x - x'\|^2)$. Hyperparameters were selected via 5-fold cross-validation: regularization parameter $C = 10.0$, $\gamma = \text{'scale'}$ ($\gamma = 1 / (n_{\text{features}} \times \text{Var}(X))$). The SVM was trained on 731 training subimages, validated on 246 samples, and tested on 243 samples. The NIR SVM achieved 97.12% test accuracy and 97.11% weighted F1-score across all seven polymer classes.

B. MIR Classifier: Random Forest (RF)

The MIR backup classifier employs a Random Forest ensemble of 200 decision trees with maximum depth 15, trained on 1,853 preprocessed FTIR spectra across six polymer classes (PET, HDPE, LDPE, PP, PS, PVC). The MIR RF model achieved 100% test accuracy and 100% weighted F1-score on the held-out test set of 398 samples, confirming the near-perfect chemical separability of these polymers in the MIR spectral window. The MIR XGBoost ablation variant (500 estimators, max depth 6, learning rate 0.05) achieved 88.7% accuracy on the three-way fine-grained task (Clear PET / Colored PET / Multi-Layer Plastic) when applied to MIR thermal imaging features.

C. Stage 1 Visual Classifier: YOLOv8-S

For binary plastic/non-plastic conveyor detection, a YOLOv8-S model was fine-tuned for 150 epochs on the Stage 1 dataset using AdamW optimizer ($\text{lr}=2 \times 10^{-4}$, cosine decay). The NIR 1D CNN (Stage 3 resin classification) trained for 120 epochs with cross-entropy loss and dropout ($p=0.3$). All training was performed on two NVIDIA RTX 4090 GPUs for the full conveyor pipeline; the core detection/classification pipeline was trained and validated on the NVIDIA RTX 4050 (6 GB VRAM).

VII. EXPERIMENTAL RESULTS AND PERFORMANCE ANALYSIS

A. Visual Pipeline Performance

The YOLOv11s detection model achieved a mAP@50 of 55.1%, a reasonable baseline given the intentional use of a lightweight model variant. The classification model substantially exceeded expectations, reaching 95.7% Top-1 accuracy and 99.7% Top-5 accuracy on ten fine-grained waste categories. Training convergence was smooth: loss fell from 2.358 at epoch 1 to 0.050 at epoch 54 (early stopping at epoch 70). Total training time was approximately 90 minutes at 8.1 iterations per second on the RTX 4050, with peak GPU memory consumption of 2.45 GB. The combined sequential inference of both models executes in under 6 ms per detected object, easily sustaining 30–60 FPS.

TABLE III. COMPLETE SYSTEM PERFORMANCE SUMMARY

Metric	Value
YOLOv11s Detection mAP@50	55.1%
YOLOv11s-cls Top-1 Accuracy	95.7%
YOLOv11s-cls Top-5 Accuracy	99.7%
Detection Inference Latency	4.5 ms/image (~222 FPS)
Classification Inference Latency	0.8 ms/image (~1,250 FPS)
Combined Pipeline Latency (per object)	<6 ms
End-to-End System Frame Rate	30–60 FPS
Peak GPU Memory (Training)	2.45 GB (RTX 4050)
NIR SVM — Standard Plastic (PVC)	89.87% confidence — Correct
NIR SVM — Black Plastic (HDPE)	15.00% confidence — Fallback Triggered
MIR RF Recovery Accuracy (Black HDPE)	99.0% — Correct
YOLOv8-S Stage 1 Accuracy (Conveyor)	96.2%
Robotic Arm Diversion Accuracy	94.8%
NIR 1D CNN Resin Accuracy	91.4%
MIR Fine-Grained Accuracy	88.7%
Overall Sorting Purity	89.2%
System Throughput (mixed stream)	18 items/minute
System Throughput (plastic-only)	26 items/minute

B. Spectral Sensing Performance

The NIR SVM classifier performed reliably on standard colored plastics: a PVC sample produced 89.87% confidence — comfortably above the 85% threshold — and was correctly classified (Fallback: False). On carbon-black HDPE, the NIR signal was severely attenuated, returning only 15.00% confidence — a 74.87 percentage-point drop that correctly triggered the MIR fallback mechanism. The MIR RF classifier subsequently identified the same black HDPE sample with 99.0% confidence (Status: SUCCESS), demonstrating the complementary spectral coverage of the two sensing modalities in practice.

Across the NIR evaluation set, the SVM achieved 97.12% accuracy and 97.11% weighted F1-score. The MIR RF achieved 100% test accuracy across all six polymer classes, with perfect precision and recall for every class. MIR was also demonstrated to resolve 91.3% of NIR low-confidence items correctly when combined NIR+MIR features were used in a late-fusion configuration.

C. NIR Resin-Type Classification by Polymer

TABLE IV. NIR STAGE 3 CLASSIFICATION RESULTS BY RESIN TYPE

Resin Type	F1 Score (%)	Resin Code	Primary Challenge
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PET	94.2	#1	Minimal
HDPE	92.8	#2	Colour variability
PVC	86.1	#3	NIR overlap with LDPE
LDPE	89.3	#4	Thin film occlusion
PP	93.7	#5	Minimal

D. Conveyor-Belt System Throughput Analysis

The integrated conveyor-belt pipeline processes 18 items per minute for a mixed waste stream, with throughput rising to 26 items per minute on a plastic-only stream. The primary bottleneck is the robotic arm cycle time of 2.7 seconds per pick. The NIR and MIR stages contribute only 6.4 ms combined latency and are not throughput-limiting at any tested stream composition. Overall sorting purity — the fraction of items in each output bin that are correctly typed — reaches 89.2%, meeting the commercial MRF purity threshold of 85–90% required for resale of recycled material streams.

Processing 18 items per minute at 89.2% purity over an 8-hour operational shift yields approximately 8,640 items sorted per day with fewer than 935 misplaced items — a significant improvement over manual sorting rates of 1,200–2,000 items per shift per worker [72], with no worker exposure to hazardous materials.

VIII. DISCUSSION

The results demonstrate that decoupling waste localization from classification — rather than relying on a monolithic end-to-end model — is a productive architectural strategy. The lightweight YOLOv11s-cls model outperforms reported accuracies of several heavier single-model approaches from the literature [4], [5] while operating at inference speeds that make real-time deployment practical on resource-constrained hardware. The sequential four-stage architecture offers a key advantage over parallel sensor fusion: each stage operates on a progressively narrower classification problem, reducing inter-class confusion and enabling per-stage model optimization.

The confidence-gated NIR→MIR fallback constitutes the most novel contribution of this work. The empirical demonstration that a 74.87 percentage-point confidence collapse on a black plastic sample reliably identifies when NIR sensing has failed — and that MIR subsequently restores accuracy to 99.0% on the same sample — validates the architectural decision to treat spectral modalities as complementary rather than interchangeable. The 85% confidence threshold represents a deliberately conservative setting that minimizes the false-negative rate for black plastic identification; future work will replace this fixed threshold with a learned, class-adaptive threshold to reduce unnecessary MIR escalations on dimly-lit but genuinely non-black plastics.

The detection mAP@50 of 55.1% represents the principal limitation of the current visual pipeline. This is attributable to the heterogeneity of real-world waste appearance, occlusion events on conveyor belts, and the deliberate choice of the small YOLOv11s model variant. Evaluation of YOLOv11m and YOLOv11l variants is expected to improve localization accuracy at the cost of increased inference latency. The webcam stage also exhibits sensitivity to dramatic lighting changes, and LDPE thin films remain the most challenging category across all sensing stages due to their deformable geometry and low NIR reflectance.

From a sustainability perspective, the proposed system directly supports United Nations Sustainable Development Goal 12.5 (substantially reduce waste generation by 2030) and SDG 14 (reduce marine plastic pollution) [93], [95]. By enabling higher-purity recycling streams, the system contributes to closing the material loop for post-consumer plastics.

IX. FUTURE WORK

Several extensions to the current system are planned for near-term development:

- **Edge Deployment:** Port the full three-modality pipeline to NVIDIA Jetson Nano or Raspberry Pi 5 for fully autonomous embedded operation without a host PC, validating real-time performance on lower-power platforms.
- **Physical Actuation Scale-Up:** Integrate multiple parallel robotic arms or high-speed delta robots to overcome the current 2.7-second arm cycle bottleneck and increase throughput toward the commercial target of 60–80 picks per minute.
- **E-Waste Extension:** Adapt the system for mobile phone component recycling streams by incorporating the Polymers Hyperspectral Imaging Dataset [16] for vision pre-training and the FTIR-Plastics dataset [28] for spectral pre-training on electronic waste polymers.

- **Adaptive Confidence Threshold:** Replace the fixed 85% NIR confidence threshold with a class-adaptive, learned threshold trained to minimize false-positive MIR escalations on dimly lit but non-black plastics, reducing unnecessary sensor activations.
- **Multi-Modal Deep Fusion:** Explore late-fusion and attention-based fusion architectures (e.g., cross-modal Swin Transformer [49]) that jointly weight RGB, NIR, and MIR modality outputs rather than the current sequential cascade, potentially improving accuracy on edge cases where no single modality is conclusive.

X. CONCLUSION

This paper presented a novel multimodal waste segregation architecture that integrates a dual-stage YOLOv11 deep learning computer vision pipeline with a confidence-gated NIR/MIR spectral sensing cascade. The vision subsystem achieves 95.7% Top-1 polymer classification accuracy with a combined inference latency under 6 ms, sustaining 30–60 FPS on edge-compatible hardware. The spectral subsystem successfully addresses the fundamental failure mode of NIR-only sorting — carbon-black plastic signal absorption — by autonomously escalating to MIR sensing, recovering classification confidence from 15.0% to 99.0% on black HDPE samples.

In full conveyor-belt deployment with a robotic sorting arm, the system achieves 96.2% binary plastic/non-plastic detection accuracy, 94.8% robotic diversion accuracy, 91.4% NIR resin-type accuracy, 88.7% MIR fine-grained accuracy, and 89.2% overall sorting purity at 18 items per minute — meeting commercial MRF purity thresholds for recyclable stream resale. Together, these contributions establish a practical, low-cost, and extensible multimodal framework for real-time waste segregation, with clear and demonstrable pathways toward industrial deployment, e-waste application, and fully autonomous embedded operation.

XI. PHYSICAL BASIS FOR SPECTRAL MODALITY SELECTION

A. NIR Spectroscopy: Operating Principles and Polymer Signatures

Near-infrared spectroscopy in the 400–1000 nm range exploits overtone and combination bands of molecular vibrations in polymer chains. PET exhibits characteristic absorption near 960 nm due to CH stretching overtones in the ethylene glycol unit. HDPE and LDPE both show strong CH₂ combination bands near 910 nm, though their crystallinity differences produce measurable baseline differences detectable by the SVM classifier. PP displays CH₃ rocking vibrations that shift key absorption maxima relative to HDPE. PVC's C–Cl bond produces unique absorption in the 900–950 nm region absent in all polyolefins, enabling high-confidence discrimination (86.1% F1) despite spectral overlap with LDPE in the 1400–1600 nm region when considering the full 400–4000 nm range.

The fundamental limitation of NIR arises from the optical properties of carbon black (CB), the most widely used black colorant in industrial plastics. CB particles are nanoscale graphitic structures (5–50 nm diameter) that absorb electromagnetic radiation across the entire ultraviolet, visible, and near-infrared spectrum through a combination of electronic absorption and scattering. Mean reflectance values below 0.08 — the system's fallback trigger threshold — are observed for CB concentrations as low as 0.5% by weight in most polymer matrices. At 2% CB concentration, NIR signal-to-noise ratios drop below the minimum required for reliable SVM classification, producing the characteristic confidence collapse observed in empirical testing (89.87% → 15.00%).

B. MIR Spectroscopy: Carbon-Black Transparency Window

The mid-infrared region (2.5–25 μm , 400–4000 cm^{-1}) provides access to fundamental molecular vibrational modes — the strongest absorbers in the infrared spectrum. Critically, carbon black's absorption cross-section decreases substantially from the NIR to the MIR: at 10 μm (1000 cm^{-1}), CB absorption is approximately two orders of magnitude lower than at 1 μm (10,000 cm^{-1}), providing a transparency window through which the underlying polymer's molecular fingerprint remains accessible.

Key diagnostic MIR absorption bands exploited by the Random Forest classifier include: C=O stretching at 1700–1750 cm^{-1} uniquely identifying PET; symmetric and asymmetric CH₂ stretching bands at 2848 and 2919 cm^{-1} present in all polyolefins (HDPE, LDPE, PP) but distinguishable by their relative intensities and accompanying CH₃ bending at 1375 cm^{-1} (PP-specific); C–Cl stretching at 600–700 cm^{-1} as a definitive PVC marker; and aromatic ring stretching at 1450–1600 cm^{-1} uniquely identifying PS. These fundamental bands are chemically invariant across polymer colors, enabling the MIR Random Forest to achieve 100% test accuracy even on the most heavily carbon-black-doped samples.

C. Confidence-Gate Calibration

The 85% confidence threshold for NIR→MIR escalation was established through receiver operating characteristic (ROC) analysis on the validation set. At 85% threshold, the false negative rate for black plastic detection (i.e., incorrectly accepting a low-quality

NIR prediction) is 0.4%, while the unnecessary MIR escalation rate on non-black plastics is 3.2%. This operating point was selected to prioritize black plastic recovery (high recall) while maintaining reasonable system throughput. The additional mean reflectance trigger (< 0.08) was added as an independent physical check, catching the rare case where a very dark non-black surface (e.g., dark green HDPE) produces low reflectance without triggering the confidence threshold.

XII. SYSTEM INTEGRATION AND DEPLOYMENT CONSIDERATIONS

A. Conveyor Belt Integration

The physical testbed comprises a 0.5 m wide conveyor belt operating at 0.25 m/s with items placed at 3-second intervals to ensure sequential one-at-a-time processing. The webcam is mounted 30 cm above the belt, capturing 1080p frames at 30 FPS. Conveyor position is tracked by an inline optical encoder (1,000 pulses/revolution) to synchronize sensor trigger timing with item location — critical for ensuring that the NIR and MIR sensors sample the correct item position without requiring physical stoppage of the belt. Belt stoppage would reduce throughput by a factor of 3–5 and is incompatible with industrial MRF deployment requirements.

Sensor placement is optimized for sequential operation: the webcam assembly is positioned at the belt entry, followed by the robotic arm station 0.8 m downstream, followed by the NIR sensor at 1.4 m, and the MIR sensor at 1.8 m from the entry point. At 0.25 m/s belt speed and 3-second item intervals, 0.75 m separates consecutive items, providing sufficient spatial separation for independent sensor acquisitions without cross-contamination between items.

B. Cost Analysis and Commercial Viability

A primary design objective of the proposed system is to minimize deployment cost relative to industrial spectroscopic sorting systems. Traditional NIR-equipped conveyor sorters (e.g., Steinert, Tomra) cost USD 150,000–500,000 per unit. The proposed architecture achieves competitive accuracy using: a consumer-grade smartphone camera (USD 200–800) as the primary optical sensor; a compact point-NIR spectrometer (USD 3,000–8,000); a compact FTIR/MIR module (USD 5,000–15,000); and a commodity GPU workstation (USD 1,500–3,000). Total per-unit hardware cost is estimated at USD 10,000–27,000 — a 5–15× cost reduction over commercial solutions while meeting commercial MRF purity thresholds. Software costs are minimized through open-source ML frameworks (PyTorch, scikit-learn, Ultralytics).

C. Scalability and Parallelization

The throughput bottleneck of 2.7 seconds per pick for the robotic arm can be addressed in production deployment by operating multiple parallel arms on the same belt — a configuration standard in industrial robotic sorting facilities. Three parallel arms at the same cycle time would increase throughput to 54 items per minute, approaching the 60–80 picks/minute reported for industrial ZenRobotics installations [73]. The NIR and MIR sensing stages add negligible latency (6.4 ms combined) and scale linearly with the number of parallel processing lanes. The deep learning inference pipeline (GPU) can process multiple items in batched inference at effectively constant latency up to batch sizes of 16–32.

D. Comparison with Prior Systems

TABLE V. COMPARISON WITH REPRESENTATIVE PRIOR SYSTEMS

System	Sensing Modalities	Black Plastic	Real-Time?	Best Reported Accuracy
TrashNet CNN [1]	RGB only	No	No	~87% (6 class)
NIR-only sorter [27]	NIR only	Fails	Yes	94%+ (no CB)
MWIR deep learning [29]	MIR only	Yes	Limited	~88% (black)
SpectralWaste [46]	RGB + HSI	Partial	No	~91% (packaging)
Wei et al. [49]	RGB + NIR	No	Partial	~93% (no CB)
Proposed System	RGB + NIR + MIR + Robotic	YES (99.0%)	YES (≤ 6 ms)	95.7% cls; 89.2% purity

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